MILITARY SPECIFICATION

TRAILER, MUNITIONS, MHU-171A/E

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of a munitions trailer designated as MHU-171A/E.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

VV-L-800	Lubricating 0i1, General Purpose, Preservative Water Displacing, Low Temperature)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Air Engineering Center, Systems Engineering and Standardization Department (Code 93), Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1740

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SPECIFICATIONS

MILITARY

MIL-L-2104	Lubricating Oil, Internal Combustion Engine, Tactical Service
* MIL-W-8611	Welding, Metal Arc and Gas, Steels, Corrosion and Heat Resistant Alloys, Process for
MIL-I-6868	Inspection process, Magnetic Particle
MIL-G-10924	Grease, Automotive and Artillery
MIL-I-45208	Inspection System Requirements

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-248	Welding and Brazing Procedure and Performance Qualification
MIL-STD-281	Automobile, Trucks, Truck-Tractors, and Trailer Dollies; Preservation and Packaging of
MIL-STD-410	Non-Destructive Testing Personnel Qualification and Certification (Eddy Current, Liquid Penetrant, Magnetic Particle, Radiographic, and Ultrasonic)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the Issues shall be those in effect on the date of the solicitation.

DOCUMENTS

* NAVSEA 0900-LP-003-8000 Surface Inspection Acceptance Standards for Metals

DRAWINGS

Naval Air Systems Command Data List DL1223AS100, Trailer, Munitions

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions, should be obtained from the contracting activity or as directed by the Contracting Activity)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of the documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN WELDING SOCIETY, INC.

* D1.1-85 Structural Welding Code, Steel

(Application for copies should be addressed to the American Welding Society Inc., 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design and construction.

3.1.1 Construction. The trailer shall be manufactured in accordance with Data List DL1223AS100 and all documents listed thereon.

3.1.2 Welding. All welding shall be in accordance with AWS DI.1-85, or MIL-M-8611 as applicable.

3.1.2.1 Welders. Welding shall be performed by personnel qualified under MIL-STD-248.

3.1.2.2 Welding Inspection. During fabrication, all welding shall be inspected in accordance with MIL-I-6868 by personnel qualified under MIL-STD-410.

* 3.1.3 Lubrication. The trailer shall be lubricated with grease MIL-G-10924, or lubricating oil MIL-L-2104, or VV-L-8OO as applicable.

3.2 Performance requirements.

* 3.2.1 Proof load. The trailer when supported by its running gear, shall withstand a proof load of 7000 $^{+100}_{-0}$ pounds.

3.2.2 Brake system.

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* 3.2.2.1 Parking brakes. The parking brakes shall be capable of holding the trailer with a 3500 $^{+10}_{-0}$ pound test load on a 11.5° grade in either direction.

* 3.2.2.2 Service brakes. The service brakes shall be capable of stopping the trailer, loaded with a 3500 $^{+100}_{-0}$ pound test load, without any tendency to jack-knife when traveling at 15 miles per hour, on level, dry concrete.

3.3 Surface inspection of metals. Non-destructive inspection shall be made of all metallic parts. This inspection shall verify that the surface conditions are in accordance with Class 1 of NAVSHIPS 0900-003-8000, surface inspection standards for metals.

3.4 Marking. Marking for shipment shall be in accordance with MIL-STD-129.

3.5 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.6 Workmanship. The trailer, including all parts and accessories, shall be fabricated in a manner that will insure that appearance, fit and adherence to specified tolerances shall be observed, and to assure within design limitations, the ability to the trailer to meet its performance requirements. Equipment having missing, inoperative, defective, bent, broken, or otherwise damaged parts will not be acceptable. The installation of hardware parts, such as catches, pins, handles, nuts, bolts, etc. shall be accomplished in such a manner as to avoid damaging the hardware or the mounting surface. Hardware or mounting surfaces damaged in this way shall be touched up to provide a continuous protective coating. Screws, nuts, and bolts showing cross threading, mutilation, detrimental or hazardous burrs shall not be acceptable. Insulated wires shall be formed into cables or ducted whenever possible. Wires, cables and brake lines shall be routed and positioned in a protective manner to avoid contact with rough or irregular surfaces and sharp edges. All welds shall be free of harmful defects such as cracks, porosity, undercuts, voids, and gaps. There shall be no evidence of burn through. Fillets shall be uniform and smooth. Angular or thickness misalignment, warpage, or dimensional change due to heat from the welding operation shall be within permitted tolerances. There shall be no damage to adjacent parts resulting from welding.

4. QUALITY ASSURANCE PROVISIONS

4.1 Repsonsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the Inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system. The contractor shall provide and maintain an inspection system in accordance with MIL-I-45208.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (4.4)

b. Quality conformance Inspection (4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with ambient environmental conditions, or as specified in the test conditions.

* 4.4 First article inspection. Each trailer shall be subjected to the tests noted in Table I, and in the sequence as shown.

TABLE I

SEQUENCE	TEST REQUIREMENT	REQ. PARA.	TEST PARA.
1	Examination of Product	3.1.1	4.6.1
2	Conformance to Drawings	3.1.1	4.6.2
3	Functioning	3.2	4.6.3
4	Service Brakes	3.2.2.2	4.6.4.2
5	Parking Brakes	3.2.2.1	4.6.4.1
6	Proof Load	3.2.1	4.6.5.1
7	Magnetic Particles Inspection	3.1.2.2	4.7

4.5 Quality conformance. Trailer components shall be inspected using sampling plans as noted on the detail drawings, or if none is specified, as directed by the procuring activity.

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4.5.1 Rejection and retest. When a unit fails to satisfy the tests noted in paragraph 4.4, no item still on hand or later produced shall be accepted until the extent and cause of failure are determined and resolved.

4.5.2 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods correcting them.

4.6. Test methods.

4.6.1 Examination of product. Each trailer shall be examined visually to determine compliance with the requirements of this specification for which tests are not specified, including materials and workmanship. Particular attention shall be given to routing protection and functioning of the lighting circuits and brake lines.

4.6.2 Conformance to drawings. Each trailer shall be Inspected to verify conformance to the engineering drawings. Dimensions and attributes not in compliance with drawing tolerances shall not be acceptable.

4.6.3 Functioning. Each trailer shall be examined to determine that wheels are correctly aligned, that all brake systems function, including back-up, and are correctly adjusted, that the electrical system is in proper working order, that all points requiring lubrication are properly lubricated, and that wheel bearings are properly adjusted and lubricated. To check alignment, trailer should be on a level surface, and with front wheels straight ahead, measure the distance between the front wheels at hub height. Mark point where measurement was made. Roll trailer forward until these marks are on the opposite side of the axle (1800). Measure the distance between the wheels at hub height at this point. The difference between the two measurements is the actual toe-in; It should not be over 1/4 inch. If toe-in is not as stated, adjust to 0 to 1/4 inch. For adjustment, loosen jam nuts on both ends of the tie-rods and rotate tie-rods in or out as required. To adjust wheel bearing, jack or prop trailer to raise wheel of the ground. Spin the wheel and tighten adjusting nut until a heavy drag is felt. Then loosen nut until wheel rotates freely and bearing end play does not exceed 0.002 to 0.003 inches. Install cotter pin through nut on axle and replace grease cap.

4.6.4 Brake system tests.

* 4.6.4.1 Parking brake tests. The trailer, loaded with a 3500 $^{+100}_{-0}$ pound test load (paragraph 4.6.5.2), shall be placed on a 11.5 degree inclined ramp, facing uphill, and with parking brakes applied, the trailer shall not roll. The loaded trailer shall then be tested facing downhill. This test should be performed after Service Brakes testing to insure that the brakes have been burnished sufficiently to achieve a successful test.

* 4.6.4.2 Service brakes. The trailer, loaded with a 3500 $^{+100}_{-0}$ pound test load (paragraph 4.6.5.2), shall then be connected to a prime mover and the trailer service brakes will then be tested in the following manner.

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* 4.6.4.2.1 Brake stopping capability test. Tow the loaded trailer at a speed of 15 mph on a level, dry concrete surface. Stop by applying brakes of towing vehicle thus actuating the trailer Inertia brake. The trailer shall exhibit no tendency to jack-knife. At the end of five test cycles, inspect the service brakes and actuating mechanism. Failure to operate satisfactorily or evidence of excessive heating or distortion shall be cause for rejection.

4.6.4.2.2 Back-up test. After acceptance under pragraph 4.6.4.2.1, the trailer shall be subjected to a back-up force. The service brakes shall hold momentarily but with continued force, shall release and allow unrestricted rearward travel. The test shall be performed five times.

4.6.5 Load tests.

* 4.6.5.1 Proof load test. The proof load testing shall be accomplished by placing a 7000 $^{+100}_{-0}$ pound test load evenly distributed between the front and rear axles of the trailer for a ten minute period. At the completion of the test, evidence of permament distortion such as warpage of trailer frame, or damage such as dents or breakage of component parts, or misalignment of wheels, or other defects detrimental to the intended end use, shall be cause for rejection.

* 4.6.5.2 3,500 pound test load. The 3500 $^{+100}_{-0}$ pound test load shall consist of bulk material such as steel or concrete evenly distributed between the front and rear axles, which shall not restrict the trailer cramping angle or drawbar movement.

* 4.7 Magnetic particle inspection test. Stressed weldments and castings, when required by the detail drawing, shall be subjected to magnetic particle inspection described in MIL-I-6868. No cracks are acceptable. Results of the magnetic particle inspection shall satisfy the criteria of NAVSEA 0900-LP-003-8000, surface inspection for metals.

4.8 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-S-28786.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be in accordance with MIL-STD-281, Level C, or as required in the contract or purchase order.

6. NOTES

6.1 Intended use. The MHU-171A/E Munition, Trailer is designed to be used with the ADU-400/E Lift Adapter for transporting and loading the TARPS (Tactical Air Reconnaissance Pod System), and certain aerial target systems.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, Number, and Date of this Specification.

b. Conditions under which Government loaned property, if any, will be made available by the Contractor.

c. Selection of applicable levels of preservation, packaging. and packing required (see paragraph 5.1)

6.3 First article. When a first article Inspection is required, the Item(s) should be a first article sample. The first article should consist of one unit(s). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who whish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to Indicate where changes (additions, deletions, modifications, corrections) from the previous issue were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies In these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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